Modular Visual Question Answering via Code Generation

A vision-language system that answers natural language questions about images. Instead of answering in one step, it breaks the question into a modular pipeline that generates Python code for visual reasoning.

How it works

1. **Image features** are extracted with a pre-trained vision-language model (**OFA** / **IDEFICS**).
2. The **question is encoded** and passed to the model with a few examples (**few-shot learning**).
3. The model **generates Python code** that performs the visual reasoning steps.
4. The code is run and its output becomes the **answer**.

Evaluation

Measured with **accuracy** and **F1-score**.

Model	Accuracy	F1-score
[add model]	[add]	[add]

Tech stack

Python, PyTorch, Hugging Face Transformers, Computer Vision, NLP, Deep Learning

Setup

```bash
git clone https://github.com/shreyadoiphode-09/[repo-name].git
cd [repo-name]
pip install -r requirements.txt
```

Run an example (replace with your own file name and arguments):

```bash
python [main_file].py --image path/to/image.jpg --question "What is on the tabl
```

Example

Add one screenshot here: an image, the question asked and the answer returned.

Known limitations

Add one or two failure cases you found and why they happened. Interviewers like to see that you checked.

Author

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